

# Solve Problems with Unit Rates

Lesson 4-7

Name: \_\_\_\_\_

Date: \_\_\_\_\_

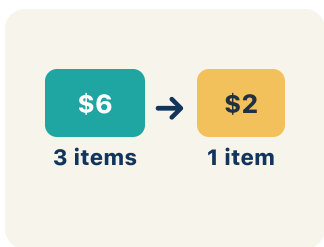
Class: \_\_\_\_\_

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## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.



*\$2.50 per pound means \$2.50 for every 1 pound*

### Unit Rate

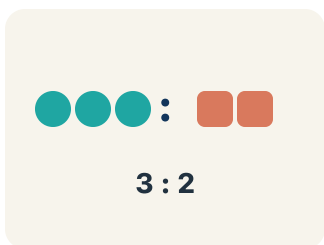
A rate for just 1 of something, like cost for 1 item.



*\$0.10 each vs \$0.15 each — the \$0.10 option is the better buy*

### Better Buy

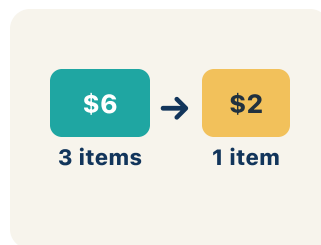
The choice that costs less for each item.



*\$3 per game < \$4 per game*

### Comparison

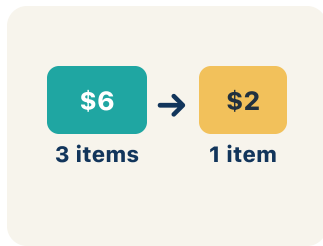
Looking at numbers to see which is bigger or smaller.



*\$0.50 per token means each token costs \$0.50*

### Per Unit

For each single item.



*120 miles in 3 hours is a rate (miles and hours are different units)*

### **Rate**

A ratio comparing two amounts with different units, like miles per hour.

A diagram illustrating a proportion. It shows a blue box with '1/2' inside, followed by an equals sign, and then another blue box with '2/4' inside.

$$5/3 = x/12 \rightarrow x = 20$$

### **Proportion**

A math sentence saying two ratios are equal. It helps find a missing number.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can solve real-world problems by using unit rates to compare options.**

#### 1 Notice

You're the new supply manager for the arcade!

#### 2 Key idea

I can solve real-world problems by using unit rates to compare options.

#### 3 Words I'll use

Unit Rate

Better Buy

Comparison

Per Unit

Rate

Proportion



#### Think About It

- How can you compare deals when the quantities are different?
- What operation helps you find the cost for just one item?
- Does the biggest package always give the best unit rate?

*See the notes in action: watch one worked all the way through, then try the next with the same steps.*



#### I do — watch

Follow each step as your teacher solves it.

**Problem:** Which is the better buy: 6 bouncy balls for \$4.50 or 10 bouncy balls for \$8.00?

- A. 6 for \$4.50 — \$0.75 each
- B. 10 for \$8.00 — \$0.80 each
- C. They cost the same per ball
- D. Not enough information

**Step 1**  $\$4.50 \div 6 = \$0.75$  each.

**Step 2**  $\$8.00 \div 10 = \$0.80$  each.

**Step 3**  $\$0.75 < \$0.80$ , so 6 for \$4.50 is the better buy.



**Answer:** A. 6 for \$4.50 — \$0.75 each

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** A faucet drips 15 ounces of water in 5 minutes. What is the unit rate?

- A. 3 ounces per minute
- B. 5 ounces per minute
- C. 15 ounces per minute
- D. 0.33 ounces per minute

**Step 1**

\_\_\_\_\_

**Step 2**

\_\_\_\_\_

**Step 3**

\_\_\_\_\_

**Answer:**

\_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** A printer prints 42 pages in 6 minutes. How many pages does it print in 15 minutes at the same rate?

- A. 105 pages
- B. 90 pages
- C. 70 pages
- D. 120 pages

Show your work:

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**As supply manager you compare token deals like 500 for \$60 and 350 for \$38.50. You can't compare totals directly, so what does finding the cost 'per unit' let you do?**

#### Level 1 support

**Start here:** Finding the cost per one token lets me compare deals fairly even when the package sizes differ.

 **Need a hint?**

**Sentence starters (optional)**

**I can't compare the totals because \_\_\_\_.**

*No puedo comparar los totales porque \_\_\_\_.*

**Finding the cost \_\_\_\_ lets me compare fairly.**

*Hallar el costo \_\_\_\_ me deja comparar de forma justa.*

**Word bank:** unit rate per unit comparison rate

**Listen for:** Students explain the packages have different quantities, so the per-unit cost (unit rate) is needed to compare deals fairly.

#### Level 2

**Does the biggest package always have the lowest unit rate? Predict, then explain how you could be tricked by package size alone.**

- The biggest package is not always cheapest per unit because \_\_\_\_.
- Package size can trick you when \_\_\_\_.

EXPLORE

**How did the unit rate help you decide the better buy between the two token suppliers in your table?**

Level 1 support

**Start here:** I found the cost per token for each supplier and chose the lower one.



**Need a hint?**

**Sentence starters (optional)**

**The unit rate for \_\_\_\_ is \$\_\_\_\_ per token.**

*La tasa unitaria de \_\_\_\_ es \$\_\_\_\_ por ficha.*

**The better buy is \_\_\_\_ because its unit rate is \_\_\_\_.**

*La mejor compra es \_\_\_\_ porque su tasa unitaria es \_\_\_\_.*

**Word bank:** unit rate better buy per unit comparison rate

**Listen for:** Students divide price by quantity for each deal (e.g., 350 for \$38.50 = \$0.11 each) and choose the supplier with the lower unit rate as the better buy.

Level 2

**One supplier had the lower price per token but a higher total. Explain why a lower unit rate, not a lower total, is what makes a deal the better buy.**

- A lower total does not mean a better buy because \_\_\_\_.
- The unit rate is what matters because \_\_\_\_.

CONNECT

**Printer A makes 150 tickets in 5 minutes and Printer B makes 200 in 8 minutes. The manager needs 600 tickets fast. How do unit rates decide the faster printer and the time needed?**

Level 1 support

**Start here:** I find tickets per minute for each printer, pick the higher rate, then divide 600 by it.

 **Need a hint?**

**Sentence starters (optional)**

**Printer A prints \_\_\_\_ tickets per minute and Printer B prints \_\_\_\_.**

*La impresora A imprime \_\_\_\_ boletos por minuto y la B imprime \_\_\_\_.*

**The manager should use Printer \_\_\_\_; 600 tickets take \_\_\_\_ minutes.**

*El gerente debe usar la impresora \_\_\_\_; 600 boletos tardan \_\_\_\_ minutos.*

**Word bank:** unit rate rate per unit comparison proportion

**Listen for:** Students find 30 and 25 tickets/min, choose Printer A (higher rate), and divide  $600 / 30 = 20$  minutes.

Level 2

**Here the higher unit rate is better, but for the token deals the LOWER unit rate was better. Explain why 'better' depends on what the rate measures.**

- A higher rate is better when \_\_\_\_, but a lower rate is better when \_\_\_\_.
- It depends on whether the rate measures \_\_\_\_ or \_\_\_\_.



## Write About the Math

### The Writing Revolution

I can explain my comparison using the words unit rate, better buy, per unit, and rate.

#### 1. Kernel Sentence subject + verb

**Model:** Unit Rate is a rate for just 1 of something, like cost for 1 item.

*Tasa unitaria es una tasa para solo 1 de algo, como el precio de 1 artículo.*

**Write a kernel sentence about unit rate. Use a subject and a verb.**

*Escribe una oración base sobre tasa unitaria. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because • but • so

**Kernel:** Unit Rate matters in math

*Tasa unitaria importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Unit Rate matters in math because \_\_\_\_.**

*Tasa unitaria importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Unit Rate matters in math, but \_\_\_\_.**

*Tasa unitaria importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Unit Rate matters in math, so \_\_\_\_.**

*Tasa unitaria importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

#### **Statement** *Afirmación*

Tell one true fact about unit rate.

*Di un hecho verdadero sobre unit rate.*

**Unit rate** \_\_\_\_.

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#### **Question** *Pregunta*

Ask a question about unit rate.

*Haz una pregunta sobre unit rate.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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#### **Exclamation** *Exclamación*

Show excitement about unit rate.

*Muestra entusiasmo sobre unit rate.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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#### **Command** *Mandato*

Tell a partner what to do with unit rate.

*Dile a un compañero qué hacer con unit rate.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**I can't compare the totals because** \_\_\_\_.

*No puedo comparar los totales porque* \_\_\_\_.

**Finding the cost** \_\_\_\_ **lets me compare fairly.**

*Hallar el costo* \_\_\_\_ *me deja comparar de forma justa.*

**The unit rate for** \_\_\_\_ **is \$** \_\_\_\_ **per token.**

*La tasa unitaria de* \_\_\_\_ *es \$* \_\_\_\_ *por ficha.*

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## Try It

Solve on your own. Check the answer key when you are done.

**1. At 55 miles per hour, how far do you travel in 4 hours?**

- A. 220 miles
- B. 59 miles
- C. 14 miles
- D. 200 miles

Show your work:

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**2. A worker earns \$54 for 6 hours. At the same rate, how much for 9 hours?**

- A. \$81
- B. \$72
- C. \$90
- D. \$60

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Two arcade supply companies offer token deals. Company X: 300 tokens for \$36. Company Y: 500 tokens for \$55. The arcade needs 1,000 tokens. Find the unit rate for each company, determine which is cheaper per token, and calculate the total cost of buying 1,000 tokens from the better deal.**

*Sentence starter: Company X charges \$\_\_\_ per token and Company Y charges \$\_\_\_ per token. The better deal is Company \_\_\_ because \_\_\_. Buying 1,000 tokens would cost \$\_\_\_.*

Show your work:

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## Reflect — Exit Ticket

**A car travels 195 miles on 6 gallons of gas. What is the unit rate in miles per gallon?**

- A. 32.5 miles per gallon
- B. 33 miles per gallon
- C. 30 miles per gallon
- D. 1,170 miles per gallon

Your answer:

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## Answer Key & Teacher Guide

1. **We Try (notes):** A. 3 ounces per minute —  $15 \div 5 = 3$  ounces per minute.
2. **You Try (notes):** A. 105 pages — Unit rate:  $42 \div 6 = 7$  pages per minute. In 15 minutes:  $7 \times 15 = 105$  pages.
3. **Try It 1:** A. 220 miles —  $55 \times 4 = 220$  miles.
4. **Try It 2:** A. \$81 —  $\$54 \div 6 = \$9$  per hour;  $9 \times \$9 = \$81$ .
5. **Exit Ticket:** A. 32.5 miles per gallon —  $195 \div 6 = 32.5$  miles per gallon.

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Unit Rate is a rate for just 1 of something, like cost for 1 item.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).